

Flight-Test Characterization and Design Implications of Hover-to-Forward-Flight Transition for a Small Quadplane VTOL Aircraft

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Abstract

Hybrid vertical take-off and landing (VTOL) configurations that combine multicopter lift with a fixed wing offer the operational flexibility of a multicopter and the cruise efficiency of an airplane, at the cost of a mode-switching transition that must remain reliably stable. This paper reports a flight-test characterization of the hover phase, the hover-to-forward-flight (MC→FW) transition, and the fixed-wing cruise of a small quadplane unmanned aerial vehicle (UAV), using telemetry recorded by an open-source PX4 autopilot, complemented by a two-dimensional Ansys Fluent computational fluid dynamics (CFD) study of the wing airfoil. Body angular rates, attitude excursions, ground speed, electrical power, bank-angle distributions, and elevator trim are extracted from the on-board estimator and cross-validated against numerical derivatives of the recorded quaternion. The aircraft exhibits a stationary hover, a smooth front transition completing at $V_T = 11.9$ m/s, peak body-axis rates that remain well within control authority, and a back transition completing in 1.86 s. The cruise window is held tightly in coordinated banked turns of approximately 50 m radius. The hover-to-cruise electrical-power ratio is $3.36\times$. A 2D Fluent angle-of-attack sweep resolves the lift

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and drag polars, identifies the onset of stall, and quantifies a $\sim 33\%$ gust-induced lift loss that anchors an operational wind ceiling. Inequality-form design rules for airspeed margin, control authority, power-system sizing, energy budget, and wind tolerance are derived from the combined empirical and CFD evidence.

Keywords: quadplane VTOL, flight test, hover-to-forward-flight transition, PX4 autopilot, computational fluid dynamics, UAV design rules, gust margin

1. Introduction

1.1. Background and motivation

Hybrid VTOL UAVs combine the runway-independent operation of a multicopter with the cruise efficiency of a fixed-wing aircraft, and have become a preferred platform for medium-range surveillance, mapping, and logistics missions [2, 3]. The development of small UAV airframes has progressed from early FPGA-based remote-piloted platforms [35] and digital multi-rotor control systems [28] toward autonomous open-source autopilot ecosystems, and wing design choices for small UAVs now span fixed cambered low-Reynolds-number sections to more sophisticated morphing-airfoil concepts [27, 33, 39]. Among hybrid configurations, the *quadplane*—a conventional airplane retrofitted with four dedicated lift rotors—offers the simplest control architecture: the multicopter (MC) and fixed-wing (FW) flight modes are mechanically decoupled, and the autopilot manages a single coordinated handover during transition. The trade-off is that the lift rotors carry parasitic mass and drag in cruise, while the wing and pusher are unused in hover, so the platform must be efficient enough in cruise to amortize the parasitic systems [4]. The test vehicle used in this study is shown at rest in Fig. 1, with the composite-skinned airframe, twin lift booms, and rear pusher motor visible.

The most demanding flight regime is the *front transition* (MC $\rightarrow$ FW), during which the wing must develop sufficient lift to support the aircraft before the lift rotors are commanded off. If transition is initiated too slowly, the wing stalls; if too quickly, the lift rotors saturate while fighting decelerating wing downwash [5, 6]. Practical PX4-class autopilots manage this trade through a single threshold airspeed V_T at which the controller commits to FW [7]. Robust transition is therefore a stability problem first—the airframe must remain trim during a regime change in which control authority transfers from rotors to aerodynamic surfaces—and a design problem second—the



Figure 1. Test vehicle at rest on the field. The airframe is a small quadplane: a fixed-wing aircraft with four boom-mounted brushless lift rotors and a single pusher propeller for fixed-wing flight. Wingspan is approximately 2.0 m and gross mass is in the 4 kg to 6 kg range.

airframe must possess sufficient power, control, and aerodynamic margins to accommodate the worst-case excursion.

1.2. Related work

The literature on small VTOL UAV transition control spans three configurations. **Tail-sitter aircraft** have dominated empirical work: Beach et al. [8] introduced a resolved tilt-twist attitude representation that avoids the Euler-angle singularities of a 90° pitch swing, Lyu et al. [9] demonstrated hierarchical attitude control on a quadrotor tail-sitter, Verling et al. [10] reported full-attitude transition on a flying-wing tail-sitter, Kubo and Suzuki [11] characterized tail-sitter transitional flight, and Ritz and D’Andrea [12] introduced a global flight-envelope controller for flying-wing tail-sitters. **Tilt-rotor and tilt-wing aircraft** have been studied by Maqsood and Go [5] on transition optimization, Geuther et al. [16] on a NASA tandem tilt-wing demonstrator, Yeh et al. [17] on an inverse machine-learning approach to tilt-wing takeoff, and May et al. [18] on tilt-wing transition strategies. **Quadplanes** have received comparatively less empirical attention: Mathur and Atkins [13] reported an experimental aerodynamic analysis of a quadplane UAS in the wind tunnel, Lee and Kim [14] proposed MPC-based longitudinal

control for a fixed-wing VTOL, Liu et al. [6] demonstrated INDI transition control on a small VTOL, and Chiwande et al. [15] presented adaptive transition control for quadplane VTOLs in urban gust environments. Survey work [2, 4, 3] consistently identifies the lift+cruise quadplane as the simplest and most energy-efficient hybrid topology for medium-range missions. The control-theoretic background for VTOL and related aerospace platforms is a parallel theme: parametric-uncertainty tracking control for VTOL vehicles has been developed in [29], distributed formation control of spacecraft and higher-order-dynamics UAVs in [36, 32], and command-filtered robust tracking on marine and surface platforms in [30], providing complementary algorithmic context for the experimental characterization reported here.

1.3. Comparison with existing methods and contributions

Three characteristics of prior empirical literature motivate the present work. *First*, many published flight-test characterizations rely on proprietary autopilot stacks whose raw telemetry is not openly accessible. *Second*, the bulk of VTOL transition work is simulation-led, with flight data appearing as a single validation point rather than the primary characterization tool. *Third*, the configuration-specific literature is heavily weighted toward tail-sitter and tilt-rotor designs, while empirical studies that report stability margins, transition timing, power budgets, and aerodynamic interpretation *jointly* on a quadplane platform are sparse.

Table 1 summarizes the scope of representative prior studies and contrasts each with the present work along eight axes: aircraft configuration, methodology, primary data source, transition-airspeed reporting, transition-duration reporting, power-budget reporting, lateral-directional characterization, and CFD aerodynamic interpretation. No prior study in the table simultaneously satisfies the joint empirical+CFD+design-rule scope on a quadplane platform.

This paper contributes the following, none of which is jointly addressed in the prior literature reviewed:

1. **Configuration focus on quadplanes.** Where prior empirical transition work largely targets tail-sitters [8, 9, 10, 12, 11] or tilt-wing and tilt-rotor airframes [5, 16, 17, 18], this paper provides a flight-test characterization of a small quadplane—a configuration that remains under-represented in the empirical literature [2, 13].
2. **Open, reproducible methodology.** The telemetry pipeline operates exclusively on the open-source PX4 `sdlog2` record. Every derived state—quaternion-to-Euler, body rates, ground speed, wind, power, energy, bank,

Table 1. Comparison of the present work with representative prior literature on small VTOL UAV transition. ✓ = reported in primary text; ✗ = not reported. Configuration codes: TS = tail-sitter; FW = flying-wing; Quad. = quadrotor. Data codes: “Emp.” = empirical (real-flight); “Sim.” = simulation; “WT” = wind tunnel.

Reference	Config.	Methodology	Data	V_T	Dur.	Pow.	Lat./dir.	CFD
Beach 2014 [8]	TS	Tilt-twist attitude	Sim.+flight	✗	✗	✗	✗	✗
Lyu 2017 [9]	Quad. TS	Hierarchical attitude	Emp.	✓	✓	✗	partial	✗
Verling 2016 [10]	FW TS	Full-attitude	Emp.	✓	✓	✗	✓	✗
Kubo 2008 [11]	TS	Transitional flight	Emp.+WT	✓	✓	✗	✗	WT
Ritz 2017 [12]	FW TS	Global envelope	Emp.	✓	✓	✗	✓	✗
Maqsood 2012 [5]	Tilt-rotor	Aerodynamic vectoring	Sim.	✓	✓	✗	✗	Invisc.
Geuther 2020 [16]	Tilt-wing	Sub-scale free-flight	Emp.	✓	partial	✗	partial	✗
Yeh 2023 [17]	Tilt-wing	Inverse-ML takeoff	Sim.	✓	✗	✗	✗	✗
May 2024 [18]	Tilt-wing	Transition strategy	Sim.	✓	✓	✗	partial	✗
Mathur 2023 [13]	QuadPlane	Wind-tunnel aerodynamics	WT	✗	✗	✗	✗	✓ (WT)
Lee 2024 [14]	FW VTOL	MPC longitudinal	Sim.	✓	✗	✗	✗	✗
Chiwande 2025 [15]	QuadPlane	INAV adaptive gust	Emp.	partial	✗	✗	partial	✗
This work	Quadplane	PX4 sdlog2 + Fluent CFD	Emp.+CFD	✓	✓	✓	✓	✓ (CFD)

turn radius, elevator trim—is cross-validated against firmware-stored fields, and per-topic CSV bundles are released (Sec. 9). The wind-tunnel work of Mathur and Atkins [13] provides aerodynamic data on a comparable platform; the present approach complements it with real-flight evidence.

- 3. Joint longitudinal and lateral-directional characterization on a single flight.** Body-rate, attitude, transition-timing, bank-angle distribution, coordinated-turn cross-check, and elevator-trim results are derived from the same flight log. No reviewed quadplane study reports all six on real flight data.
- 4. CFD aerodynamic interpretation with gust anchoring.** A 2D Ansys Fluent angle-of-attack sweep gives $C_L(\alpha)$ and $C_D(\alpha)$ polars, multi-AoA pressure contours, three-dimensional flow visualization, and a quantitative gust-perturbation case that converts the empirical V_T threshold into a physically anchored, gust-margin-quantified design quantity.
- 5. Inequality-form design rules.** Airspeed margin, control authority, power-system sizing, energy budget, and operational wind ceiling are translated from the combined empirical and CFD evidence into compact design inequalities (Sec. 8).

The results derive from a single flight on one airframe and constitute a case study; broader statistical validation through additional flights and platforms is identified as the central future-work theme (Sec. 9).

The remainder of the paper is organized as follows. Section 2 presents

the mathematical preliminaries. Section 3 describes the data-extraction and CFD methodologies, together with the aerodynamic and energetic relations specific to the quadplane configuration. Sections 4 and 5 describe the test vehicle and the experimental setup. Section 6 reports the flight-test results. Section 7 presents the computational aerodynamic analysis. Section 8 extracts the design implications, and Section 9 concludes.

2. Preliminaries

This section establishes the mathematical framework used to derive aircraft states from the recorded PX4 telemetry and from the complementary CFD sweep.

2.1. Nomenclature

The symbols used throughout this paper are listed in Table 2.

Table 2. Nomenclature.

Symbol	Description	Symbol	Description
ϕ, θ, ψ	roll, pitch, yaw (rad/deg)	L, D, T	aerodynamic lift, drag, thrust (N)
p, q, r	body angular rates (rad/s, deg/s)	C_L, C_D, C_M	lift, drag, pitching-moment coeff.
q_w, q_x, q_y, q_z	unit attitude quaternion components	C_{L_α}	lift-curve slope (per rad)
V_N, V_E, V_D	NED-frame ground velocity (m/s)	α	angle of attack (rad or deg)
V_h	horizontal ground speed (m/s)	n	load factor
$V_{\text{air}}, \text{IAS}$	indicated airspeed from pitot (m/s)	W, g	aircraft weight (N), gravity (m/s ²)
$V_T, V_{T,\text{back}}$	front/back transition airspeeds (m/s)	ρ, μ	air density (kg/m ³), dyn. visc. (Pa·s)
$h, \dot{h}$	altitude AGL, climb rate (m, m/s)	S, c, b	wing area, chord, span (m ² , m, m)
$\delta_e, \delta_a, \delta_r$	elevator, aileron, rudder (deg/PWM)	Re	Reynolds number
R	turn radius (m)	$\eta_{\text{prop}}, \eta_{\text{ESC}}$	propeller, ESC efficiency
P, E	electrical power (W), energy (J or Wh)	N_r	number of lift rotors
V_{bat}, I	battery pack voltage (V), current (A)	A_d	single-rotor disc area (m ²)
		σ	standard deviation

2.2. Attitude representation

The aircraft attitude is stored on board as a unit quaternion $\mathbf{q} = (q_w, q_x, q_y, q_z)$ with $q_w^2 + q_x^2 + q_y^2 + q_z^2 = 1$. The corresponding Tait–Bryan Euler angles (yaw–pitch–roll, Z - Y - X intrinsic rotation) follow [24, 25, 26]

$$\phi = \text{atan2}(2(q_w q_x + q_y q_z), 1 - 2(q_x^2 + q_y^2)), \quad (1)$$

$$\theta = \arcsin[2(q_w q_y - q_z q_x)], \quad (2)$$

$$\psi = \text{atan2}(2(q_w q_z + q_x q_y), 1 - 2(q_y^2 + q_z^2)). \quad (3)$$

2.3. Body angular-rate kinematics

The relationship between body-axis angular rates (p, q, r) and Euler-angle rates is [24, 26]

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}, \quad (4)$$

which is valid in the non-singular range $|\theta| < \pi/2$.

2.4. Ground speed, airspeed, and wind

The horizontal ground speed is $V_h = \sqrt{V_N^2 + V_E^2}$ and the climb rate is $\dot{h} = -V_D$. In a stationary hover ($V_h \approx 0$), the steady wind component along the aircraft body- x axis is

$$V_{\text{wind}} \approx V_{\text{air}} - V_h. \quad (5)$$

2.5. Electrical power, energy, and endurance

Instantaneous electrical power, accumulated energy, and pure-phase endurance are

$$P(t) = V_{\text{bat}}(t) I(t), \quad E = \int_{t_0}^{t_1} P(t) dt, \quad t_{\text{phase}} = \frac{E_{\text{pack}}}{\bar{P}_{\text{phase}}}, \quad (6)$$

and the mission-average power for a hover-duty fraction η_h is

$$\bar{P}_{\text{mission}} = \eta_h \bar{P}_{\text{hover}} + (1 - \eta_h) \bar{P}_{\text{cruise}}. \quad (7)$$

2.6. Coordinated turn and load factor

In a coordinated turn at airspeed V and bank angle ϕ , the horizontal force balance yields [23, 25]

$$R = \frac{V^2}{g \tan \phi}, \quad \dot{\psi} = \frac{g \tan \phi}{V}, \quad n = \frac{1}{\cos \phi}, \quad (8)$$

with n the load factor. The turn radius R can be cross-checked kinematically from the observed yaw rate as $R = V_h/|\dot{\psi}|$; agreement of the two estimates indicates negligible side-slip.

2.7. Elevator trim

For a standard servo with neutral PWM 1500 μs and full deflection $\pm\delta_{\max}$ at $\pm 500 \mu\text{s}$, the commanded elevator angle is

$$\delta_e [\text{deg}] = \frac{\text{PWM} - 1500}{500} \delta_{\max}. \quad (9)$$

2.8. Aerodynamic forces and Reynolds number

Lift, drag, and pitching moment on a wing of reference area S and chord c at airspeed V are

$$L = \frac{1}{2}\rho V^2 S C_L(\alpha), \quad D = \frac{1}{2}\rho V^2 S C_D(\alpha), \quad M = \frac{1}{2}\rho V^2 S c C_M(\alpha). \quad (10)$$

In the linear regime, $C_L(\alpha) \approx C_{L_0} + C_{L_\alpha}\alpha$ with $C_{L_\alpha} = dC_L/d\alpha$ the lift-curve slope. The drag polar follows the parabolic approximation

$$C_D(\alpha) \approx C_{D_0} + k C_L^2(\alpha), \quad (11)$$

with C_{D_0} the zero-lift drag coefficient and k the induced-drag factor. The chord Reynolds number is

$$Re = \frac{\rho V c}{\mu}, \quad (12)$$

and the lift-to-drag ratio is

$$\frac{L}{D} = \frac{C_L(\alpha)}{C_D(\alpha)}, \quad (13)$$

maximized when $dC_D/dC_L = C_D/C_L$. Low-Reynolds-number effects on $C_{L,\max}$ and drag are reviewed in [20, 21, 22].

3. Methodology

This section describes the methodology in four parts: the binary-log data-extraction pipeline (Sec. 3.1); the numerical operators that produce the results of Sec. 6 (Sec. 3.2); the aerodynamic and energetic balance equations that connect the flight measurements to the airframe’s design quantities (Sec. 3.3); and the Ansys Fluent setup used for the complementary aerodynamic sweep of Sec. 7 (Sec. 3.4).

3.1. Data extraction from the sdlog2 binary format

The PX4 binary log used in this study is in the legacy `sdlog2` format, identified by the file-leading magic bytes `0xA3 0x95 0x80`. Each message is preceded by a three-byte header (`0xA3 0x95 <MSG_TYPE>`); the `FMT` message (`<MSG_TYPE>=0x80`) declares the binary layout of all subsequent message types. The file is parsed using `sdlog2_dump.py` from the PX4-Autopilot v1.7.0 release tag, which produces per-topic CSV outputs filtered with the `-m` option. Global time alignment is performed by forward-filling the `TIME_StartTime` field, giving every row a microsecond-precision wall-time coordinate.

The following topics are used: `ATT` (attitude quaternion and body rates, ~ 250 Hz); `ATSP` (attitude setpoint); `AIRS` (calibrated indicated airspeed, ~ 1 kHz); `BATT` (filtered pack voltage and current); `LPOS` and `LPSP` (local position estimator and setpoint); `GPOS` (global position); `OUT0` and `OUT1` (PWM commands); and `VTOL` (mode flags). Euler angles obtained from Eqs. (1)–(3) are verified against the firmware-stored `ATT_Pitch`, `ATT_Roll`, and `ATT_Yaw` fields to residuals below $2 \times 10^{-6}^\circ$, and body rates from `ATT_*Rate` are cross-validated against numerical Euler-angle derivatives obtained via Eq. (4).

3.2. Numerical operators on per-topic samples

The CSV samples are non-uniform. Numerical derivatives are computed by centred differences,

$$\dot{x}_i \approx \frac{x_{i+1} - x_{i-1}}{t_{i+1} - t_{i-1}}. \quad (14)$$

Phase-window integrals (Eq. (6)) are evaluated by the trapezoidal rule,

$$E \approx \sum_{i=0}^{N-1} \frac{1}{2} (P(t_i) + P(t_{i+1})) (t_{i+1} - t_i). \quad (15)$$

Per-phase statistics use the unbiased sample estimators

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \sigma_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2. \quad (16)$$

A state is reported as “essentially zero in the mean” when $|\bar{x}| < \sigma_x$.

Phase windows are identified using the recorded rotor-wing and transition flags. Let $f_{\text{Rw}}(t)$ and $f_{\text{Tr}}(t)$ denote the rotor-wing and transition indicators.

The phase logic is

$$\text{phase}(t) = \begin{cases} \text{MC hover} & f_{\text{Rw}} = 1, f_{\text{Tr}} = 0, V_h < 0.5 \text{ m/s}, \\ \text{MC} \rightarrow \text{FW transition} & f_{\text{Tr}} = 1, \dot{f}_{\text{Rw}} < 0, \\ \text{FW cruise} & f_{\text{Rw}} = 0, f_{\text{Tr}} = 0, \\ \text{FW} \rightarrow \text{MC transition} & f_{\text{Tr}} = 1, \dot{f}_{\text{Rw}} > 0, \end{cases} \quad (17)$$

with the MC $\rightarrow$ FW and FW $\rightarrow$ MC branches distinguished by the sign of the rotor-wing edge. The cruise window for elevator-trim analysis is further restricted to samples satisfying the level-flight criterion $|\phi| < 5^\circ$ and $|\dot{h}| < 0.5 \text{ m/s}$.

3.3. Aerodynamic and energetic balance specific to the quadplane

For a quadplane operating at the autopilot-held cruise airspeed V_c , four balance equations connect the empirical measurements of Sec. 6 to the airframe's design quantities. In level steady cruise, the lift balances the weight,

$$\frac{1}{2}\rho V_c^2 S C_L^{\text{cruise}} = W \implies C_L^{\text{cruise}} = \frac{2W}{\rho V_c^2 S}, \quad (18)$$

and the pusher thrust balances the drag,

$$T_{\text{pusher}} = D = \frac{1}{2}\rho V_c^2 S C_D^{\text{cruise}}. \quad (19)$$

The corresponding electrical power required at the pusher motor, accounting for ESC and propeller efficiency, is

$$P_{\text{cruise,req}} = \frac{T_{\text{pusher}} V_c}{\eta_{\text{prop}} \eta_{\text{ESC}}} = \frac{\rho V_c^3 S C_D^{\text{cruise}}}{2 \eta_{\text{prop}} \eta_{\text{ESC}}}. \quad (20)$$

Hence the measured cruise power $\bar{P}_{\text{cruise}}$ from Eq. (6) fixes the product $S C_D^{\text{cruise}}/(\eta_{\text{prop}}\eta_{\text{ESC}})$ and, given a nominal S , allows the aerodynamic and electromechanical loss contributions to be separated.

In hover, the four lift rotors collectively support the weight. The per-rotor ideal hover thrust, the disc loading, and the ideal induced power per rotor are

$$T_{\text{per}} = \frac{W}{N_r}, \quad \text{DL} = \frac{T_{\text{per}}}{A_d}, \quad P_{\text{ideal}}^{(1)} = T_{\text{per}} \sqrt{\frac{T_{\text{per}}}{2\rho A_d}}, \quad (21)$$

and the figure-of-merit relating ideal induced power to measured electrical power per rotor is

$$\text{FoM} = \frac{P_{\text{ideal}}^{(1)}}{P_{\text{measured}}^{(1)}} = \frac{T_{\text{per}} \sqrt{T_{\text{per}}/(2\rho A_d)}}{\bar{P}_{\text{hover}}/N_r}. \quad (22)$$

A measured $\bar{P}_{\text{hover}}$ together with a known A_d therefore yields a configuration-level efficiency metric directly comparable across airframes.

In a coordinated banked turn at bank angle ϕ , the wing must support the load factor $n = 1/\cos \phi$ from Eq. (8), requiring an increased lift coefficient

$$C_L^{\text{turn}}(\phi) = \frac{2nW}{\rho V^2 S} = \frac{C_L^{\text{cruise}}}{\cos \phi}. \quad (23)$$

The corresponding stall speed in the turn rises as

$$V_{\text{stall}}^{\text{turn}}(\phi) = V_{\text{stall,1g}} \sqrt{n} = V_{\text{stall,1g}} / \sqrt{\cos \phi}, \quad (24)$$

which bounds the maximum bank that can be sustained at the cruise airspeed.

For a head-on gust of magnitude Δv that reduces the freestream from V to $V - \Delta v$ at fixed angle of attack, the lift drops as the square of velocity,

$$\frac{\Delta L}{L} = \left(1 - \frac{\Delta v}{V}\right)^2 - 1 \approx -\frac{2\Delta v}{V} \quad \text{for } \Delta v \ll V. \quad (25)$$

The corresponding airspeed margin for stall avoidance at the transition airspeed V_T is

$$\Delta v_{\text{max}} = V_c - V_T, \quad (26)$$

so the operational head-wind ceiling satisfies $V_{\text{wind,max}} \lesssim \Delta v_{\text{max}}$. The CFD case of Sec. 7.4 provides the high- α correction to Eq. (25), where boundary-layer evolution amplifies the linear V^2 scaling.

3.4. Computational aerodynamic setup

A complementary 2D CFD study is performed on a section of the wing airfoil using Ansys Fluent (2022 R2 / 2023 R1) under an academic licence. The simulation is steady and incompressible, with a pressure-based solver and the SST $k\text{-}\omega$ turbulence model of Menter [19], which is appropriate for attached and mildly separated airfoil flow at moderate Reynolds number. The freestream condition is $V_\infty = 15 \text{ m/s}$ (the autopilot-held cruise airspeed; see Sec. 6), $\rho = 1.225 \text{ kg/m}^3$, and $\mu = 1.789 \times 10^{-5} \text{ Pa}\cdot\text{s}$, with turbulence intensity 1% at the inlet. Eq. (12) gives chord Reynolds numbers in the low-Reynolds-number range typical of small UAV applications [20, 21]. A symmetric C-grid mesh extends approximately 10 chord lengths upstream and 20 chord lengths downstream of the airfoil, with $y^+ \lesssim 1$ on the airfoil surface for boundary-layer resolution. A grid-refinement study comparing coarse, medium, and fine meshes at the cruise condition confirmed that the

integrated lift and drag coefficients change by less than 1% between the medium and fine meshes; the medium mesh is used for the production angle-of-attack sweep. An angle-of-attack sweep is performed from $\alpha = -5^\circ$ to $+20^\circ$ in 5° increments, and lift and drag are extracted from CFD-Post via the global-direction force calculator on the wing surface. A separate gust-perturbation case applies a 2 m/s freestream perturbation at $\alpha = +15^\circ$ to estimate gust-induced lift loss per Eq. (25). Convergence is monitored on the scaled residuals (each $< 10^{-5}$) and on the lift and drag force time histories.

4. Hardware and equipment

The test vehicle is a small quadplane VTOL UAV with a high-mounted, high-aspect-ratio main wing and an H-tail empennage on twin booms (Fig. 2). Four electrically driven lift rotors are mounted on the boom centrelines, one rotor at the front and one at the rear of each boom, giving a conventional “+”/“X” quadcopter arrangement when viewed from above. A separate brushless pusher motor with a folding propeller is mounted at the rear of the central fuselage and provides all thrust during forward flight. Control surfaces include differential ailerons on the main wing, an elevator on the horizontal stabilizer, and rudders on the twin vertical fins. The airframe is hand-built in fibreglass/composite skin over a foam core and is powered by a 6-cell lithium-polymer battery pack. The wing planform and airfoil cross-section are shown in Fig. 3; the airfoil is a thin, moderately cambered low-Reynolds-number section typical of small UAV applications [20, 21].

The principal system specifications are summarized in Table 3 and the equipment inventory in Table 4.

5. Experimental setup

The flight analyzed in this paper was a ~ 24 min session conducted on 4 September 2022 in an open agricultural field near Dhaka, Bangladesh (23.85° N, 90.51° E, near sea level), of which ~ 6.5 min was armed (Fig. 4). Ambient conditions were tropical with hazy visibility and a moderate steady wind subsequently estimated via Eq. (5). The mission consisted of: (i) arming and pre-flight hover; (ii) vertical climb to ~ 45 m AGL; (iii) one complete MC $\rightarrow$ FW front transition; (iv) ~ 4 min of fixed-wing cruise in a circular loiter pattern at ~ 45 m AGL and ~ 15.2 m/s airspeed; (v) one complete FW $\rightarrow$ MC back transition; and (vi) vertical descent and recovery. A complete time-tagged log (log_5_2022-9-4-18-02-56.px4log) was recovered from the au-



Figure 2. Test vehicle in operation. (a) Planform from below showing the wing, H-tail empennage and four lift rotors. (b) Closer in-flight view. (c) Banked-turn profile. (d) Head-on silhouette. (e) Wide test field with operator and aircraft visible. (f) Post-landing on the field. (g) Underside planform view in steady cruise, with all four lift rotors visible and the H-tail at the rear of the twin booms. (h) Side three-quarter view in late transition, with the wing developing lift authority. (i) Low-altitude pass over the recovery area with the ground operator visible. The ground-rest view is in Fig. 1.

Table 3. System specifications of the test quadplane. Values marked [†] are estimated from imagery and component sizes; remaining values are derived from the flight log of Sec. 5.

Category	Parameter	Value
Configuration	Aircraft class	Hybrid quadplane VTOL (lift+cruise)
	Lift propulsion	4× brushless DC, boom-mounted
	Cruise propulsion	1× brushless DC pusher (rear fuselage)
	Control surfaces	2 aileron, 1 elevator, 2 rudder
Geometry	Wing configuration	High-mounted, high aspect ratio
	Empennage	H-tail on twin booms
	Wingspan b	~ 2.0 m [†]
	Wing chord c	~ 0.25 m [†]
	Reference area S	~ 0.5 m ^{2†} (estimated from photogrammetry)
	Fuselage length	~ 1.7 m [†]
Mass / Loading	Gross mass m	~ 4 kg to 6 kg [†]
	Wing loading W/S	~ 80 N/m ² to 120 N/m ^{2†}
	Hover disc loading $W/(N_r A_d)$	~ 30 N/m ² to 50 N/m ^{2†}
Performance (measured)	Cruise airspeed V_c	15.19(59) m/s
	Front-transition airspeed V_T	11.92 m/s
	Back-transition airspeed $V_{T,\text{back}}$	15.62 m/s
	Front-transition duration	4.53 s
	Back-transition duration	1.86 s
	Typical bank in turn	$ \phi \approx 25^\circ$
Electrical	Battery	6S LiPo, ~ 10 A h, ~ 222 W h
	Mean hover power	1269(24) W
	Mean cruise power	378(131) W
	Peak transition power	1597 W (~ 4 s duration)
Endurance / Range	Pure-hover endurance (theoretical)	~ 10.5 min
	Pure-cruise endurance (theoretical)	~ 35 min
	Practical cruise range	~ 20 km
Autopilot	Flight stack	PX4 v1.7.0 (open-source)
	Log format	PX4 legacy sdlog2 (binary)

Table 4. Equipment inventory of the test airframe. Items marked [‡] are typical of PX4-compatible builds of this class and reflect the as-built configuration; the published telemetry is independent of the specific manufacturer where not stated otherwise.

Subsystem	Component	Specification / Description
Avionics	Flight controller	Pixhawk-class, PX4-compatible flight controller
	IMU	MEMS 3-axis accelerometer + gyro [‡]
	Magnetometer	3-axis digital magnetometer [‡]
	Barometer	Digital pressure altimeter [‡]
	GPS receiver	u-blox-class L1 with integrated compass [‡]
	Airspeed sensor	Pitot tube + differential pressure transducer
Communications	Telemetry	SiK-compatible 433 MHz radio modem [‡]
	RC receiver	SBUS / PPM [‡]
	Ground station	QGroundControl (PX4 ecosystem)
Lift propulsion ($\times 4$)	Motor	Brushless DC outrunner, ~ 600 kV to 900 kV [‡]
	Speed controller	BLHeli ESC, ≥ 25 A continuous [‡]
	Propeller	Fixed-pitch, ~ 10 – 12 in diameter [‡]
Cruise propulsion ($\times 1$)	Motor	Brushless DC, ~ 700 kV to 900 kV [‡]
	Speed controller	BLHeli ESC, ≥ 30 A continuous [‡]
	Propeller	Folding, ~ 10 – 13 in diameter [‡]
Actuation	Aileron servos ($\times 2$)	Standard hobby-grade, ~ 0.10 s/ 60° [‡]
	Elevator servo ($\times 1$)	Standard hobby-grade [‡]
	Rudder servos ($\times 2$)	Standard hobby-grade [‡]
Power	Battery pack	6S LiPo, nominal 22.2 V, ~ 10 A h
	Power module	PX4-compatible $V+I$ sensor, in-line
Airframe	Wing	Fibreglass/composite skin over foam core
	Fuselage	Fibreglass/composite shell
	Booms	Twin carbon-fibre booms with $4\times$ lift rotors

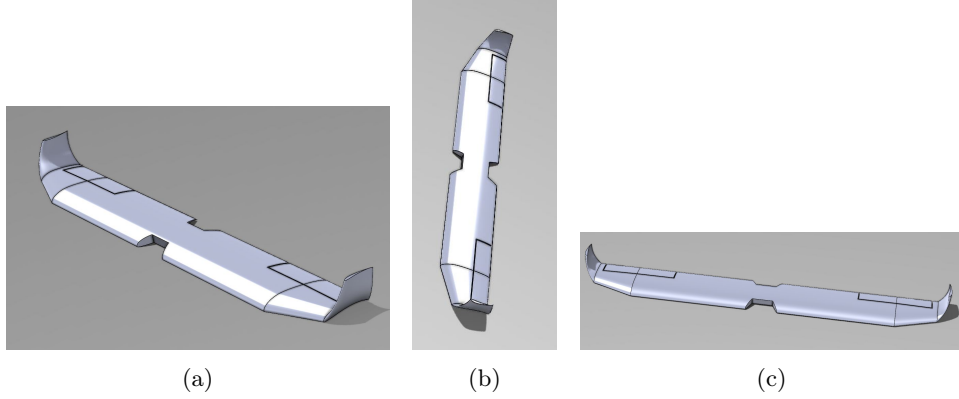


Figure 3. Wing geometry. (a) Isometric high-angle CAD view of the full wing assembly showing the canted tip surfaces and the central trailing-edge cut-out at the fuselage interface. (b) Near end-on view illustrating planform thickness and the tip-cant angle. (c) Lower three-quarter perspective showing the swept-up tip surfaces and the inboard/outboard control-surface outlines on the wing.

topilot SD card; the per-topic CSV bundle is released alongside this paper (Sec. 9). An overview of the entire armed flight is given in Fig. 5.



Figure 4. Test environment. Open field site with palm trees and distant structures; operator at right with ground-control station; aircraft visible in low hover during recovery.

6. Flight-test results

This section reports the results of the pipeline of Sec. 3 applied to the recorded flight log: hover (Sec. 6.1), transition (Sec. 6.2), power and en-

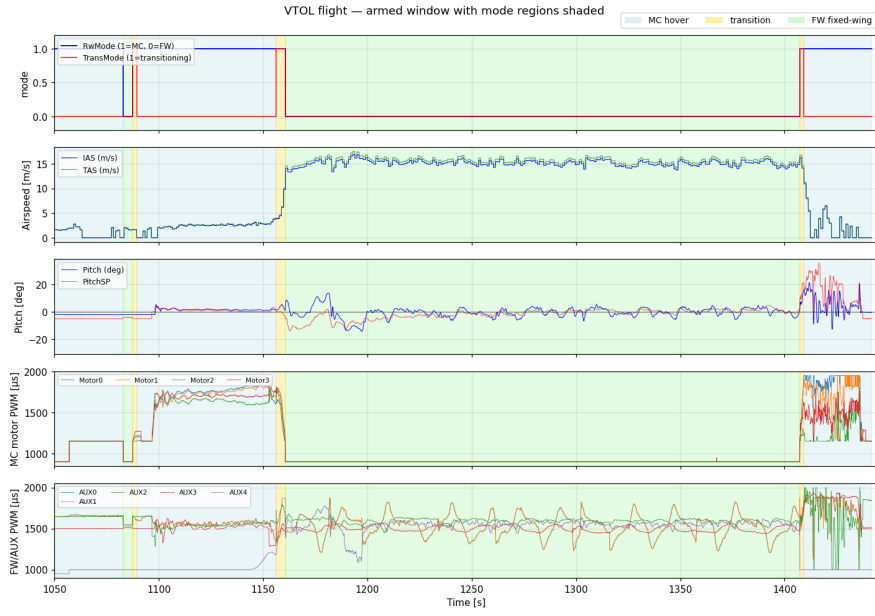


Figure 5. Armed-flight overview. From top: VTOL mode flags (RwMode, TransMode); pitot airspeed (IAS) and true airspeed (TAS); pitch angle θ and pitch setpoint θ_{cmd} ; multicopter lift-motor PWM commands; fixed-wing/auxiliary PWM commands including the pusher. Shaded regions denote MC hover (blue), transition (gold), and FW cruise (green).

ergy (Sec. 6.3), lateral-directional cruise (Sec. 6.4), and longitudinal trim (Sec. 6.5). The aerodynamic interpretation appears in its own section (Sec. 7).

6.1. Hover-phase stability

Figure 6 shows p, q, r over a window spanning the last ~ 11 s of hover, the 4.5 s MC $\rightarrow$ FW transition, and the first ~ 9 s of cruise. The hover rates are tightly bounded:

$$\bar{p}_{\text{hover}} = +0.51 \pm 6.29 \text{ deg/s}, \quad \bar{q}_{\text{hover}} = -0.18 \pm 3.65 \text{ deg/s}, \quad \bar{r}_{\text{hover}} = -5.91 \pm 8.34 \text{ deg/s}.$$

Mean roll and pitch rates are within sensor noise of zero; the small yaw rate is consistent with deliberate pre-transition heading alignment. The horizontal ground-speed magnitude V_h is essentially zero (0.16 ± 0.07 m/s), while the pitot reads $V_{\text{air}} = 2.93 \pm 0.34$ m/s (Fig. 7); Eq. (5) gives a wind component along body- x of ~ 2.77 m/s.

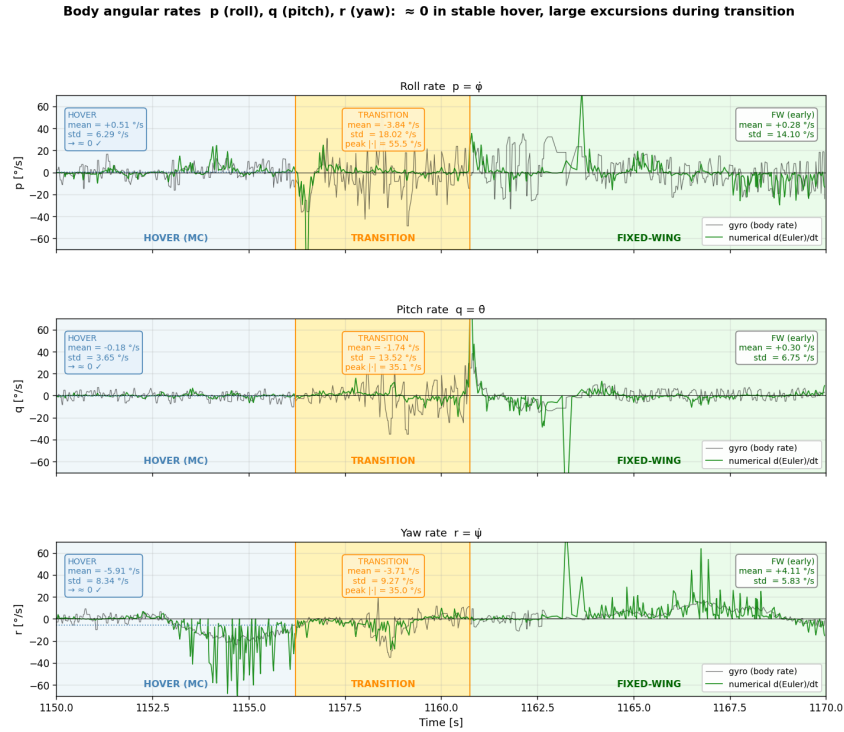


Figure 6. Body-axis angular rates p, q, r across hover $\rightarrow$ transition $\rightarrow$ fixed-wing. Gray: gyro rates; green: numerical derivatives of Euler angles, Eq. (4). Hover rates near-zero; transition peaks $\sim 55^\circ/\text{s}$ (roll), $\sim 35^\circ/\text{s}$ (pitch/yaw).

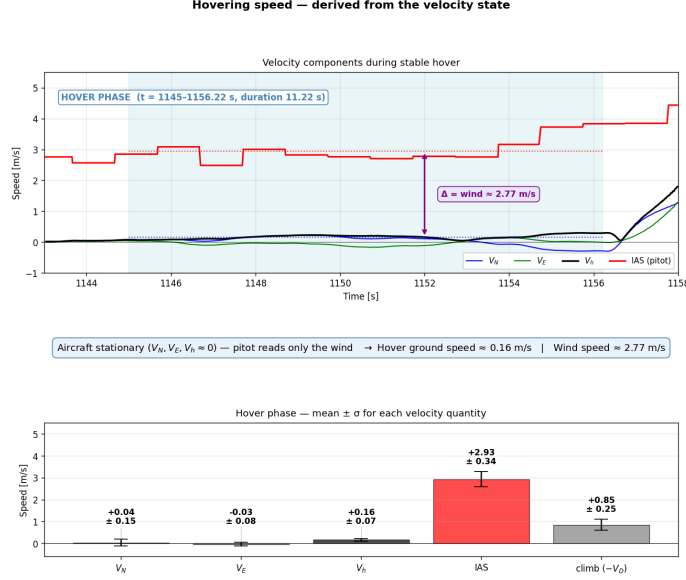


Figure 7. Velocity components during stable hover. NED components and V_h hug zero, while the pitot reads $2.93 \pm 0.34 \text{ m/s}$; the difference is the steady wind along body- x ($\sim 2.77 \text{ m/s}$).

6.2. Hover-to-forward-flight transition

The MC→FW transition begins when the autopilot toggles **TransMode** from 0 to 1 and ends when the lift rotors are commanded off. The pitot airspeed at each toggle defines the transition airspeed thresholds (Table 5, Fig. 8).

The front transition starts well below stall ($V \approx 3.83 \text{ m/s}$), with the pusher motor spooling up while the lift rotors maintain support, and completes at $V_T = 11.92 \text{ m/s}$ ($\sim 28\%$ margin to cruise airspeed). The back-transition trigger ($V_{T,\text{back}} = 15.62 \text{ m/s}$) is slightly above cruise, so the lift rotors begin spooling up while the wing still has positive lift authority.

Pitch behaviour during transition (Fig. 9) is a primary indicator of how the airframe converts thrust into airspeed. From $t = 1156$ – 1158 s , IAS climbs from 3.8 to $\sim 6 \text{ m/s}$ as the pusher spools up; pitch rises modestly to $+2^\circ$. Between 1158 and 1159 s the airspeed accelerates through the wing’s low-speed regime and pitch peaks at $\theta_{\text{max}} = +5.40^\circ$. From 1159 to 1160.75 s the wing develops authority, the elevator actively trims, and pitch settles to the cruise mean of -0.57° . Transition body-rate peaks are $|p|_{\text{max}} \approx 55.5^\circ/\text{s}$, $|q|_{\text{max}} \approx 35.1^\circ/\text{s}$, $|r|_{\text{max}} \approx 35.0^\circ/\text{s}$ (Fig. 6, gold region), well below the PX4

Table 5. Transition airspeed thresholds and durations from the TransMode flag toggles.

Event	Time [s]	IAS [m/s]
Front-transition commit	1156.22	3.83
Front-transition complete	1160.75	11.92 ($= V_T$)
Back-transition commit	1407.32	15.62 ($= V_{T,\text{back}}$)
Back-transition complete	1409.18	12.88
<i>Front-transition duration</i>	—	4.53 s
<i>Back-transition duration</i>	—	1.86 s

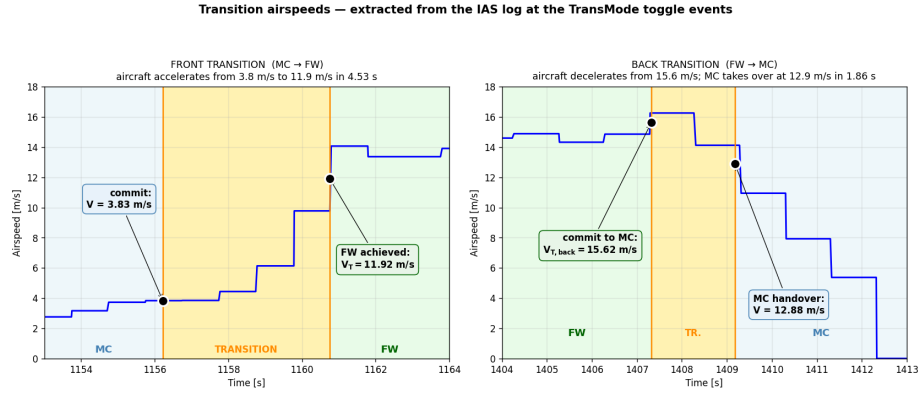


Figure 8. Pitot IAS through the front (left) and back (right) transitions. Black dots mark TransMode toggles; $V_T = 11.92$ m/s is the minimum airspeed for wing-borne flight on this airframe.

default saturation limit ($\sim 220^\circ/\text{s}$). The back transition completes in 1.86 s with body rates $\leq 20^\circ/\text{s}$ throughout.

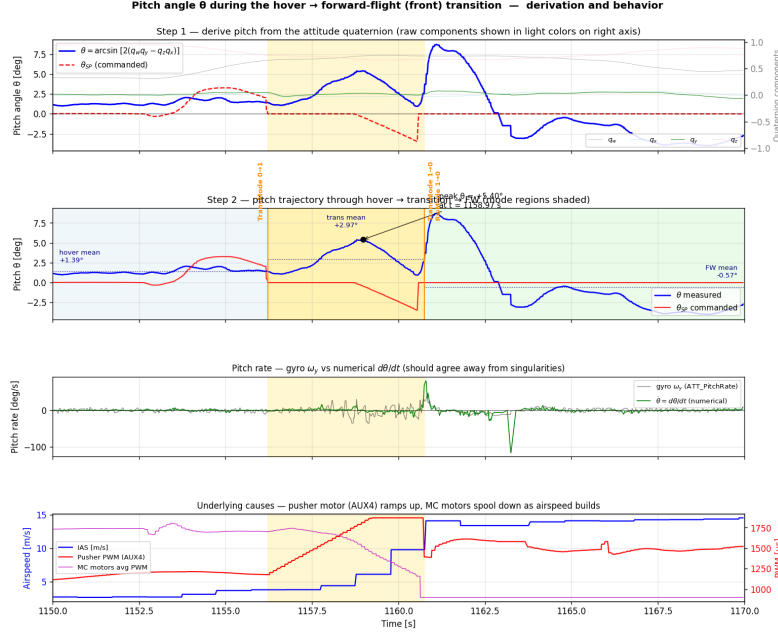


Figure 9. Pitch behaviour during front transition. Top-left: derivation from quaternion, Eq. (2). Top-right: pitch trajectory with phase means; peak $+5.4^\circ$ at $t = 1158.97\text{ s}$. Bottom-left: pitch rate (gyro and numerical derivative). Bottom-right: pusher PWM and airspeed rise.

6.3. Power and energy budget

Phase-resolved electrical power is summarized in Table 6 and Fig. 10.

The hover-to-cruise power ratio is $\bar{P}_{\text{hover}}/\bar{P}_{\text{cruise}} = 1269/378 \approx 3.36$, the front-transition peak is 1597 W sustained for $\sim 4\text{ s}$ (sets ESC/battery peak), and the avionics baseline is 97 W ($\sim 25\%$ of cruise). Applying Eq. (6) with $E_{\text{pack}} \approx 223\text{ Wh}$ gives a pure-hover endurance $\sim 10.5\text{ min}$ and a pure-cruise endurance $\sim 35\text{ min}$ (theoretical range $\sim 31\text{ km}$). A practical mission with 30% pack reserve and $\sim 8\text{ Wh}$ hover overhead yields $\sim 20\text{ km}$ of usable cruise range. Hover power is remarkably steady ($\sigma_P/\bar{P} \approx 1.9\%$). Applying Eq. (20) with the measured $\bar{P}_{\text{cruise}} \approx 378\text{ W}$ and an assumed $\eta_{\text{prop}}\eta_{\text{ESC}} \approx 0.45$ gives an aerodynamic $T_{\text{pusher}} \approx 11.3\text{ N}$ and an L/D ratio at cruise $\approx W/T_{\text{pusher}}$ in the range 3.5–5.0 for the 4–6 kg mass band, consistent with low-Re cambered airfoils [20, 21]. The measured pack-level power profile sets the requirements

Table 6. Phase-resolved electrical power and energy.

Phase	Duration [s]	Mean P [W]	Peak P [W]	Energy [Wh]
Pre-takeoff (avionics)	27	97 ± 12	130	0.73
Climb-out (MC)	50	1010 ± 407	1244	14.01
Stable hover (MC)	11	1269 ± 24	1333	3.92
Front transition (MC→FW)	4.5	1368 ± 146	1597	1.73
Cruise (FW)	235	378 ± 131	1010	24.81
Back transition (FW→MC)	1.9	380 ± 13	393	0.20
Landing (MC)	31	890 ± 253	1245	7.71

on the on-board energy-management layer, for which data-driven state-of-charge estimation and balancing schemes [31, 38] are an active research direction.

6.4. Lateral-directional cruise behaviour

The 235 s cruise window is a continuous circular loiter at ~ 45 m AGL. The autopilot holds IAS to 15.19 ± 0.59 m/s (4% of setpoint) and altitude to ± 2 m. Figure 11 shows the bank-angle trace and distribution: the aircraft spends 83.8% of cruise turning ($|\phi| > 10^\circ$), with mean right-turn bank $+27.1^\circ$, mean left-turn bank -23.1° , and peaks of $+53.8^\circ/-45.6^\circ$. The mean signed bank $+1.03^\circ$ confirms left/right cancellation over the loiter. The turn radius is computed two ways via Eq. (8): the aerodynamic formula and the kinematic $R = V_h/|\dot{\psi}|$. Median radii of $R_{\text{formula}} = 54$ m and $R_{\text{kinematic}} = 48$ m agree to $\sim 10\%$, confirming coordinated turning. Typical operating radius is ~ 50 m, with tight ~ 20 m turns at maximum bank. Eq. (23) requires $C_L^{\text{turn}}/C_L^{\text{cruise}} = 1/\cos 25^\circ \approx 1.10$ at the typical bank; from Fig. 14, this remains well within the linear regime.

6.5. Longitudinal trim during cruise

Figure 12 shows the elevator PWM command during cruise, with red dots marking samples satisfying the level-flight criterion ($|\phi| < 5^\circ$, $|\dot{h}| < 0.5$ m/s; $n = 65$ samples). The level-cruise mean is $1607 \mu\text{s}$ with $\sigma = 11.9 \mu\text{s}$, an offset of $+107 \mu\text{s}$ from neutral. Applying Eq. (9) with $\delta_{\text{max}} = 30^\circ$ yields $|\delta_e^{\text{trim}}| \approx 6.4^\circ$ with sub-degree jitter, consistent with a well-tuned pitch-attitude loop.

7. Computational aerodynamic analysis

To interpret the empirical V_T threshold and quantify the wing’s sensitivity to airspeed disturbance, an angle-of-attack sweep on the wing airfoil was performed in Ansys Fluent following the setup of Sec. 3.4. This

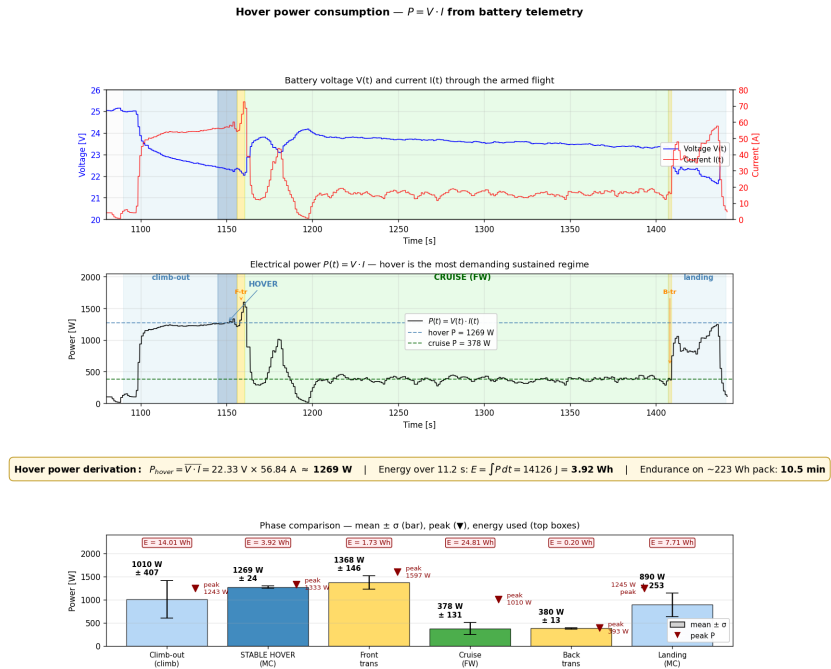


Figure 10. Electrical power budget. Top: $V(t)$, $I(t)$. Middle: $P(t) = V \cdot I$ with hover/cruise means. Bottom: per-phase mean and peak power and energy.

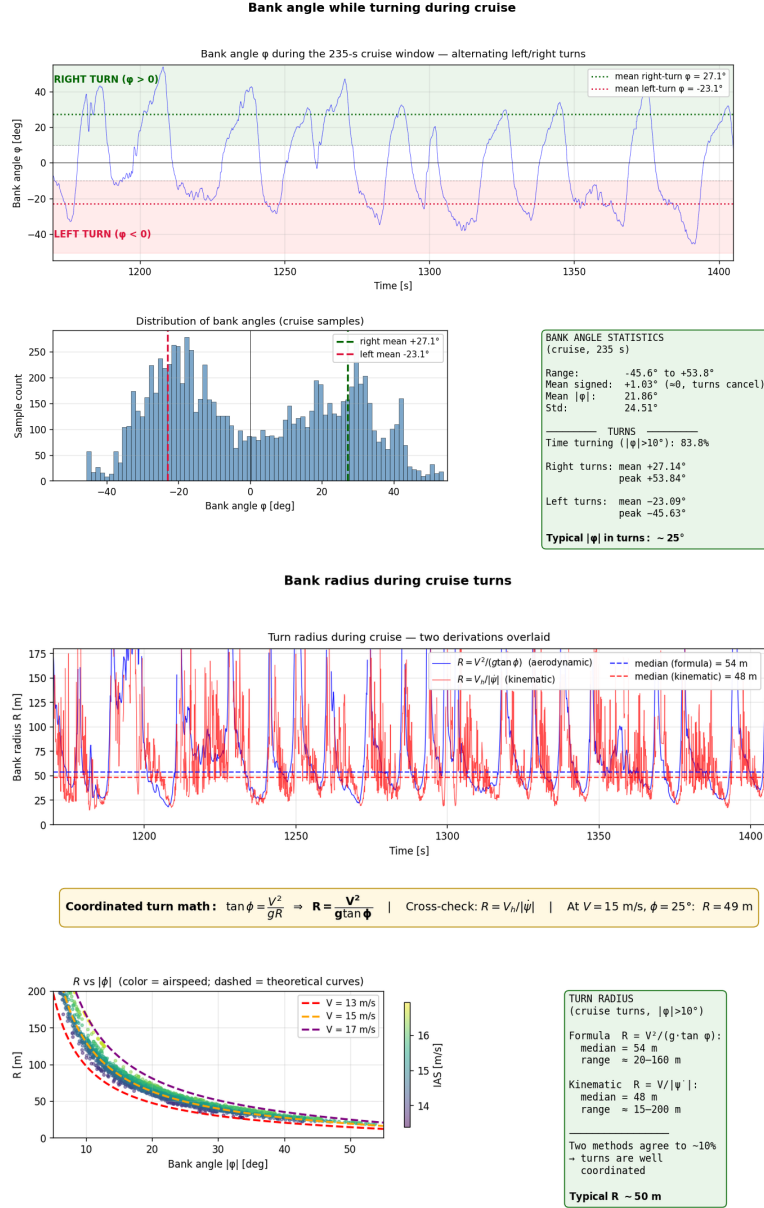


Figure 11. Lateral-directional cruise. Top: bank-angle ϕ trace and distribution; typical $|\phi| \approx 25^\circ$, time-fraction turning 83.8%. Bottom: turn radius from $R = V^2/(g \tan \phi)$ (blue) and $R = V_h/|\dot{\psi}|$ (red); two-method agreement to $\sim 10\%$ confirms coordinated turns. Bottom-right scatter: R vs $|\phi|$ falls on theoretical curves for $V \in [13, 17]$ m/s.

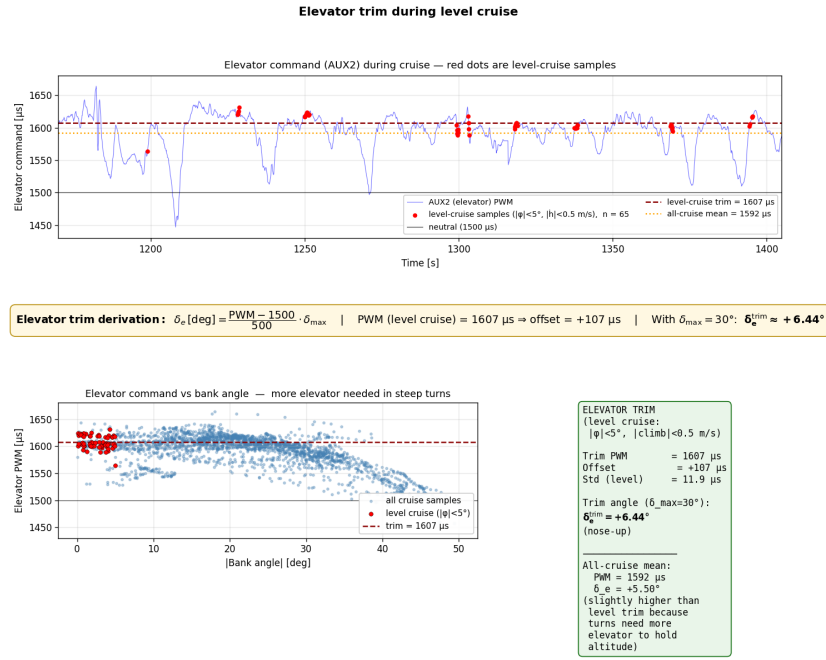


Figure 12. Elevator command (AUX2) during cruise. Red dots are level-cruise samples ($|\phi| < 5^\circ$, $|\dot{h}| < 0.5 \text{ m/s}$, $n = 65$). Level-cruise PWM mean 1607 μs (offset +107 μs); $|\delta_e^{\text{trim}}| \approx 6.4^\circ$.

section reports the multi-AoA pressure field (Sec. 7.1), the lift and drag polars (Sec. 7.2), qualitative 3D flow visualization (Sec. 7.3), and the gust-perturbation case (Sec. 7.4). Implications for V_T and the wind ceiling are summarized in Sec. 7.5.

7.1. Multi-angle pressure field

Figure 13 shows the static pressure contour around the airfoil at $\alpha = -5^\circ, +5^\circ, +15^\circ, +20^\circ$, $V_\infty = 15$ m/s. At $\alpha = -5^\circ$ the pressure field is nearly fore-aft symmetric with weak upper-surface low pressure, an essentially zero-lift condition for the cambered section. At $\alpha = +5^\circ$ a clear low-pressure suction peak develops on the upper surface near the leading edge and a higher-pressure plateau forms on the lower surface, the canonical attached-flow configuration. At $\alpha = +15^\circ$ the suction peak intensifies and migrates toward the leading edge. At $\alpha = +20^\circ$ the suction peak concentrates within a small leading-edge region while the lower-surface pressure plateau is at its strongest; the topology signals that the upper-surface boundary layer is on the edge of separation.

7.2. Lift and drag polars

Figure 14 reports the dimensionless $C_L(\alpha)$ and $C_D(\alpha)$ obtained by integrating surface pressures via Eq. (10). The lift coefficient is essentially linear from $\alpha = -5^\circ$ to $\alpha = +15^\circ$ with fitted lift-curve slope $C_{L_\alpha} \approx 1.56/\text{rad}$ ($\approx 0.027/\text{deg}$), consistent with low- Re thin-cambered airfoils [20, 21]; the slope is below the inviscid $2\pi/\text{rad}$ value because of low- Re viscous effects characteristic of small UAV airfoils [22]. The departure from linearity between $+15^\circ$ and $+20^\circ$ marks the onset of stall. The drag curve grows quadratically with α , fitted as $C_D(\alpha) \approx C_{D_0} + k\alpha^2$, consistent with the parabolic polar of Eq. (11).

7.3. Three-dimensional flow visualization

Qualitative 3D flow visualization (Ansys Discovery, lower-fidelity preview solver) is shown in Fig. 15. Streamlines and velocity vectors confirm largely attached flow on both surfaces, with mild streamline curvature in the immediate vicinity of the wing. The 3D views also indicate that spanwise flow is small under symmetric inflow, motivating the use of the 2D Fluent sweep above as a representative section model. Higher-fidelity 3D simulations are reserved for future work.

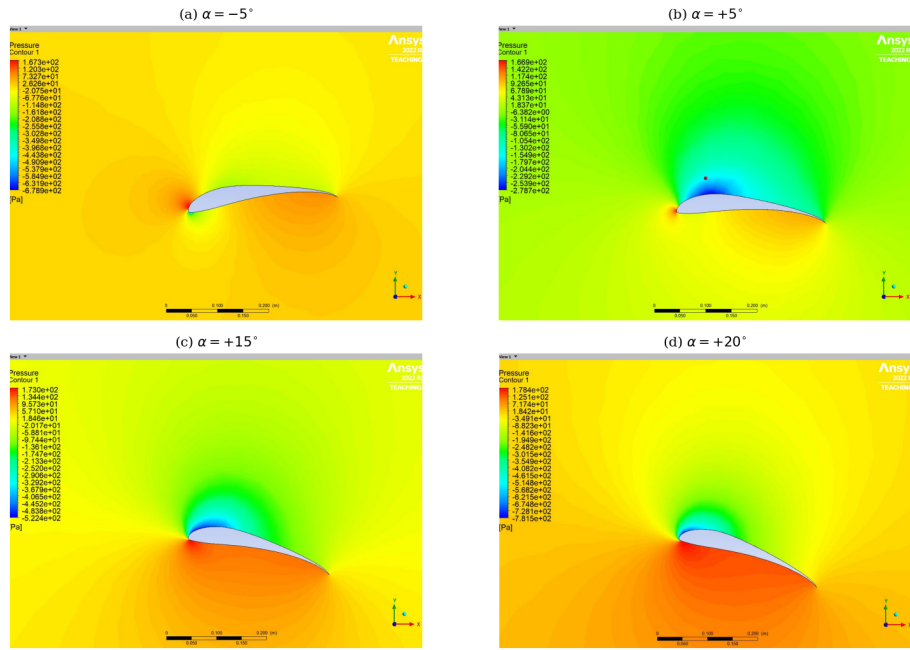


Figure 13. Static pressure contour around the wing airfoil at four angles of attack, $V_\infty = 15 \text{ m/s}$. (a) $\alpha = -5^\circ$: near-symmetric field. (b) $\alpha = +5^\circ$: working attached flow. (c) $\alpha = +15^\circ$: intensified suction peak. (d) $\alpha = +20^\circ$: leading-edge-concentrated suction, near-stall topology.

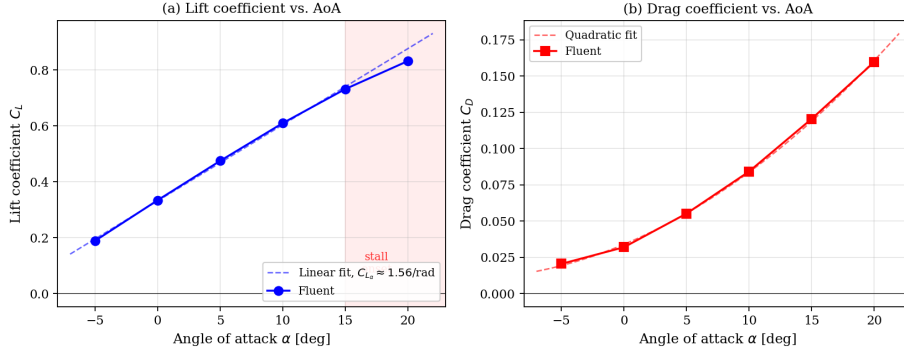


Figure 14. Wing airfoil polars. (a) $C_L(\alpha)$: linear regime from -5° to $+15^\circ$ with fitted slope $C_{L_\alpha} \approx 1.56/\text{rad}$, flattening above $+15^\circ$ (red band) marks stall onset. (b) $C_D(\alpha)$: quadratic growth.

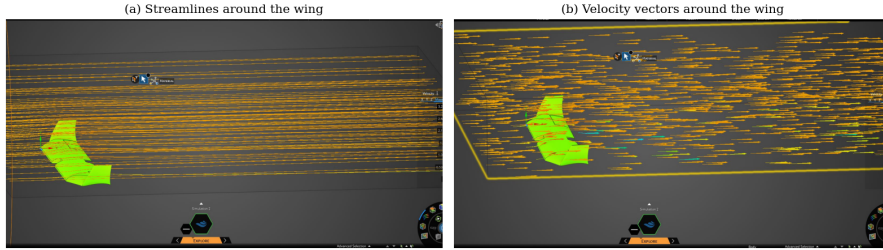


Figure 15. Preliminary 3D flow visualization (Ansys Discovery), $V_\infty \approx 2.5 \text{ m/s}$. (a) Streamlines: largely attached flow with mild curvature near the wing. (b) Velocity vectors: predominantly streamwise, small spanwise component.

7.4. Gust-perturbation case

A separate Fluent case applies a 2 m/s freestream perturbation at $\alpha = +15^\circ$, simulating a head-on gust encountered while the wing operates at high-AoA transition margin. The integrated lift drops from 50.4 N (steady) to 33.8 N (gusted), a $\sim 33\%$ reduction (Fig. 16). Linear Eq. (25) predicts $2\Delta v/V = 4/15 \approx 27\%$ from dynamic-pressure scaling alone; the additional $\sim 6\%$ drop reflects boundary-layer evolution on the suction side near the stall edge, consistent with the leading-edge-concentrated topology of Fig. 13(c). This case anchors the operational wind ceiling derived in Sec. 8.5.

7.5. Aerodynamic interpretation of V_T

At $V_\infty = V_T = 11.92 \text{ m/s}$ and the measured transition pitch $\theta \approx +3^\circ$, the lift balance of Eq. (18) requires a C_L^{trans} a factor $(15/11.92)^2 \approx 1.58$

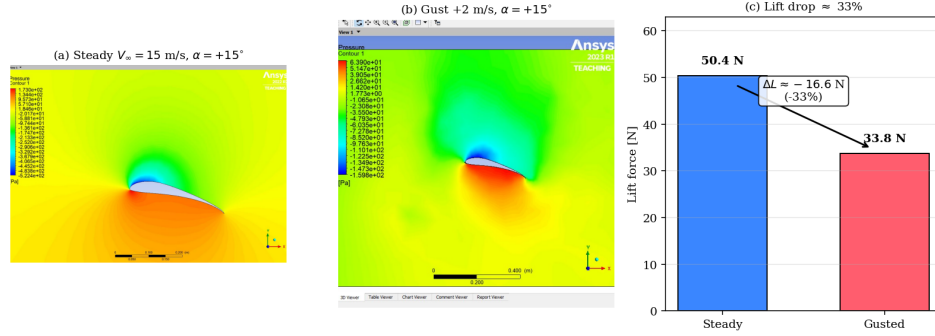


Figure 16. Gust-perturbation case at $\alpha = +15^\circ$. (a) Steady $V_\infty = 15$ m/s pressure contour. (b) Perturbed $V_\infty - 2$ m/s pressure contour. (c) Integrated lift drops by $\sim 33\%$, anchoring the wind-margin rule.

above the cruise C_L . From Fig. 14(a) this required C_L lies in the linear regime well below the stall ceiling at $\alpha \approx +15^\circ$. A reduction in V_T would push the wing closer to stall; an increase would waste energy on prolonged transition. The gust case (Sec. 7.4) shows that a modest 2 m/s disturbance consumes much of this margin at high α , providing the quantitative basis for the wind-tolerance rule of Sec. 8.5.

8. Design implications

The measured flight characteristics and CFD interpretation of V_T motivate several design rules for quadplane-class VTOL UAVs, stated in inequality form for reusability, with the caveat that the present results derive from a single flight on one airframe; broader statistical validation is future work.

8.1. Transition airspeed margin

With $V_T = 11.92$ m/s sitting $\sim 28\%$ below cruise (15.2 m/s), the CFD lift curve places the wing well within the linear $C_L(\alpha)$ regime at V_T . The measurement and CFD together support

$$V_{\text{cruise}} \geq 1.25 V_T \quad \text{and} \quad V_{T,\text{back}} \geq V_{\text{cruise}}. \quad (27)$$

8.2. Control authority budget

Peak transition body rates of $\sim 55^\circ/\text{s}$ (roll) and $\sim 35^\circ/\text{s}$ (pitch/yaw) sit well below the PX4 default $220^\circ/\text{s}$ limits, giving more than $4\times$ headroom:

$$\dot{\Phi}_{\text{controller,max}} \geq 4 \dot{\Phi}_{\text{trans,peak}}. \quad (28)$$

8.3. Power-system sizing

Hover (~ 1.3 kW sustained) and front transition (1.6 kW transient) define worst-case discharge. For a 223 Wh 6S pack (~ 10 Ah at 22 V), continuous 60 A implies 6C and peak 72 A implies 7.2C—within commodity 6S LiPo specifications. Per-rotor hover current of ~ 15 A indicates ≥ 25 A ESCs are adequate.

8.4. Energy budget and range

With Eq. (7), missions with $\eta_{\text{hover}} \leq 0.10$ are cruise-dominated and approach the ~ 35 min pure-cruise endurance. With 30% pack reserve and ~ 8 Wh hover overhead, practical cruise range is ~ 20 km. The avionics baseline of 97 W ($\sim 25\%$ of cruise) is itself a candidate for reduction.

8.5. Operational wind ceiling

The Fluent gust case (Sec. 7.4) shows a 2 m/s airspeed disturbance at $\alpha = +15^\circ$ reduces lift by $\sim 33\%$. With Eq. (26), a conservative operational ceiling is

$$V_{\text{wind,max}} \lesssim V_{\text{cruise}} - V_T \approx 3.3 \text{ m/s}, \quad (29)$$

or larger only with confirmed gust-rejection authority. Adaptive transition strategies in gust-heavy environments are an active research direction [15].

8.6. Configuration choice

The lift+cruise layout offers two advantages consistent with comparative analyses [4, 2, 3]: mode independence (lift rotors decoupled from pusher and wing, eliminating tilt-mechanism complexity [18, 16]) and clean stall margins (all four lift rotors idle in cruise). The trade-off is that the lift rotors and booms are parasitic in cruise, observable in the cruise/hover ratio.

9. Conclusions

Flight-test telemetry from a small quadplane VTOL UAV, recorded by an open-source PX4 autopilot in the legacy `sdlog2` format, has been combined with a 2D Ansys Fluent angle-of-attack sweep to characterize the hover phase, the MC $\rightarrow$ FW transition, the FW cruise, and the FW $\rightarrow$ MC back transition. The empirical pipeline is reproducible from the binary log; the CFD provides the physical interpretation of the empirical V_T . A comparison table (Table 1) confirms that no prior study covers the same combination of axes (configuration, methodology, data source, transition-airspeed reporting, transition-duration reporting, power-budget reporting, lateral-directional characterization, and CFD aerodynamic interpretation) on a quadplane platform.

In hover, body rates are within sensor noise ($|\dot{p}|, |\dot{q}| < 0.5^\circ/\text{s}$) and pitot recovers the steady wind (2.77 m/s). The MC→FW transition completes in 4.53 s at $V_T = 11.92$ m/s with pitch bounded at $+5.4^\circ$ and body rates peaking at $\sim 55^\circ/\text{s}$; the back transition completes in 1.86 s. Cruise airspeed is held within 4% of setpoint with $\sim 25^\circ$ banks in coordinated turns of ~ 50 m radius, two derivations agreeing to 10%. Elevator trim is $\delta_e^{\text{trim}} \approx +6.4^\circ$. The hover-to-cruise power ratio is $3.36\times$; transition surges to 1.6 kW.

The CFD sweep provides $C_L(\alpha)$ and $C_D(\alpha)$ polars, identifies stall onset at $\alpha \gtrsim +15^\circ$, and demonstrates a 2 m/s gust-induced lift loss of $\sim 33\%$ —supplying the quantitative basis for the operational wind ceiling. Design implications follow directly from the combined evidence.

The results derive from a single flight on one airframe; broader statistical validation through additional flights and platforms is required and is the central theme of future work. Planned extensions include higher-fidelity 3D CFD with confirmed reference area and mass, IMU vibration spectra during cruise vs. hover, per-motor current isolation, and disc-loading/figure-of-merit estimation once rotor geometry is measured directly. Further methodological extensions include the integration of safe model-free reinforcement-learning controllers with state and control constraints via control barrier functions [37] and discrete-event safety supervision over the autopilot’s mode logic [34].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Md Azizul Islam: Investigation, Methodology (hardware), Resources (airframe build and integration), Validation, Visualization, Writing – original draft. **Md Nur-A-Adam Dony:** Conceptualization, Formal analysis, Methodology (data analysis and CFD), Software, Visualization, Writing – original draft, Writing – review & editing, Project administration. **Mehedi Hasan:** Investigation, Validation, Writing – review & editing. **S. G. M. Hossain:** Funding acquisition (aircraft), Resources (aircraft), Validation, Writing – review & editing.

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Data availability

The analysis pipeline operates on per-topic CSV exports of the original PX4 `sdlog2` binary log (`log_5_2022-9-4-18-02-56.px4log`, 4 September 2022). The fifteen CSV topic files used in this work (ATT, ATSP, ARSP, AIRS, BATT, GPOS, GPS, LPOS, LPSP, OUT0, OUT1, PWR, RC, STAT, VTOL), the original binary log, and the Python notebooks used to generate the figures of this paper are released at the public GitHub repository

<https://github.com/AdamDony/quadplane-px4-flight-data>

and a corresponding archival snapshot is deposited on Zenodo; the Zenodo DOI will be inserted upon acceptance.

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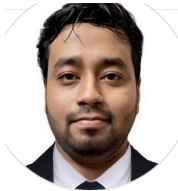
Vitae



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